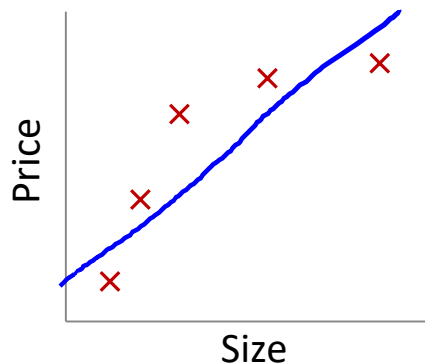


Regularization

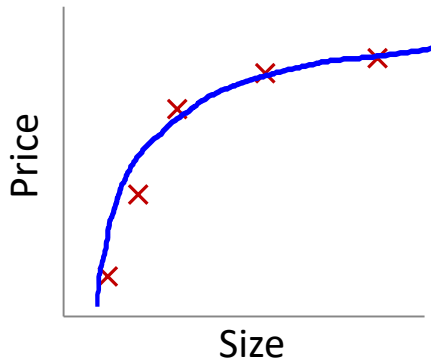
The problem of
overfitting

Example: Linear regression (housing prices)



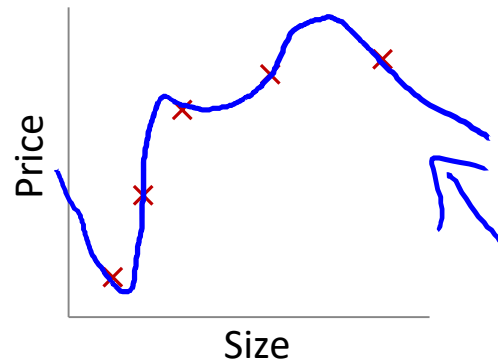
$$\rightarrow \theta_0 + \theta_1 x$$

"Underfit" "High bias"



$$\rightarrow \theta_0 + \theta_1 x + \theta_2 x^2$$

"Just right"

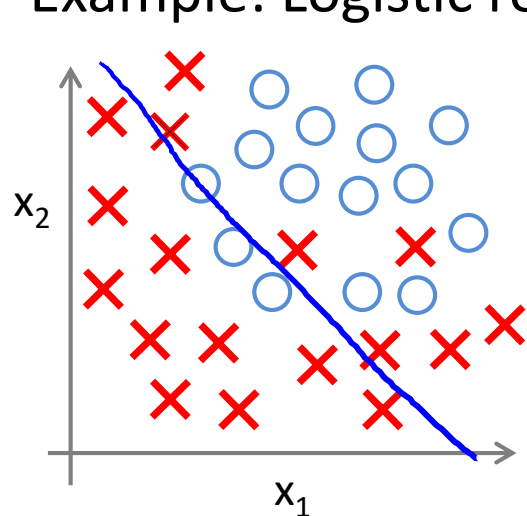


$$\rightarrow \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$$

"Overfit" "High variance"

Overfitting: If we have too many features, the learned hypothesis may fit the training set very well ($J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 \approx 0$), but fail to generalize to new examples (predict prices on new examples).

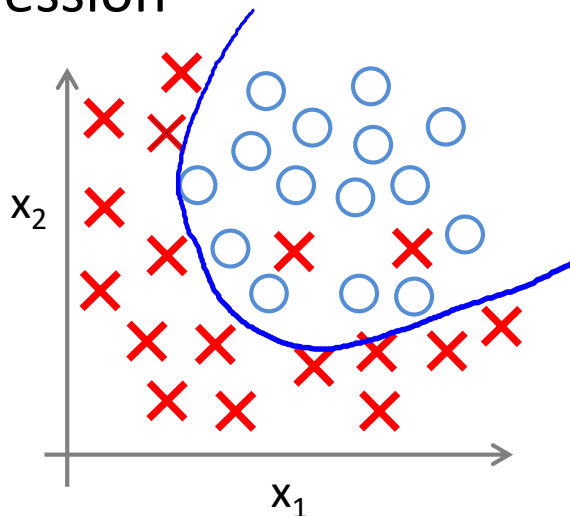
Example: Logistic regression



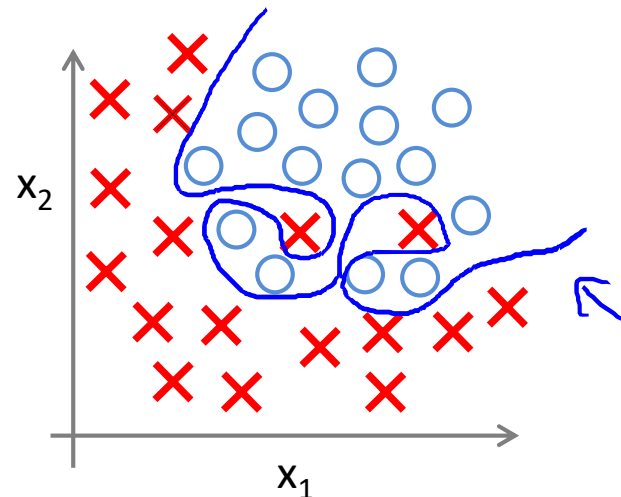
$$\rightarrow h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

(g = sigmoid function)

"Underfit"



$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 + \theta_5 x_1 x_2)$$



$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^2 x_2 + \theta_4 x_1^2 x_2^2 + \theta_5 x_1^2 x_2^3 + \theta_6 x_1^3 x_2 + \dots)$$

"Overfit"

Addressing overfitting:

x_1 = size of house

x_2 = no. of bedrooms

x_3 = no. of floors

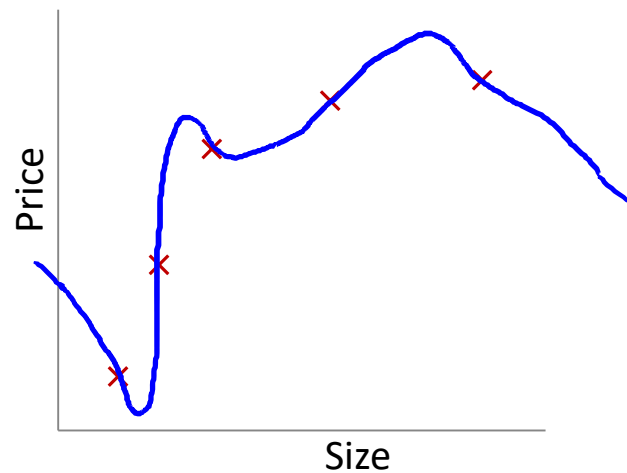
x_4 = age of house

x_5 = average income in neighborhood

x_6 = kitchen size

$\vdots$

x_{100}



Addressing overfitting:

Options:

1. Reduce number of features.

→ — Manually select which features to keep.

→ — Model selection algorithm (later in course).

2. Regularization.

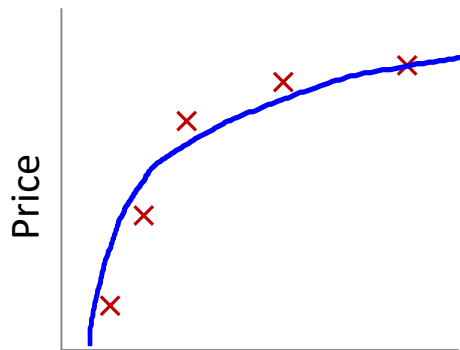
→ — Keep all the features, but reduce magnitude/values of parameters θ_j .

— Works well when we have a lot of features, each of which contributes a bit to predicting y .

Regularization

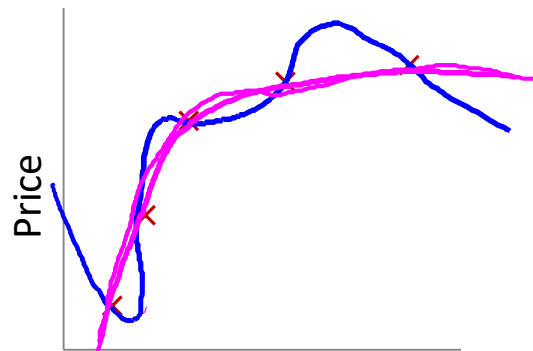
Cost function

Intuition



Size of house

$$\theta_0 + \theta_1 x + \theta_2 x^2$$



Size of house

$$\theta_0 + \theta_1 x + \theta_2 x^2 + \cancel{\theta_3 x^3} + \cancel{\theta_4 x^4}$$

Two pink arrows point upwards from the θ_3 and θ_4 terms in the equation, indicating they are being penalized or set to zero.

Suppose we penalize and make θ_3, θ_4 really small.

$$\rightarrow \min_{\theta} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + 1000 \theta_3^2 + 1000 \theta_4^2$$

Below the equation, two blue expressions are written and underlined with pink lines:

$$\theta_3 \approx 0 \quad \theta_4 \approx 0$$

Regularization.

Small values for parameters $\theta_0, \theta_1, \dots, \theta_n$

- “Simpler” hypothesis
- Less prone to overfitting

$$\rightarrow \boxed{\theta_3, \theta_4} \approx 0$$

Housing:

- Features: $x_1, x_2, \dots, x_{100}$
- Parameters: $\theta_0, \theta_1, \theta_2, \dots, \theta_{100}$

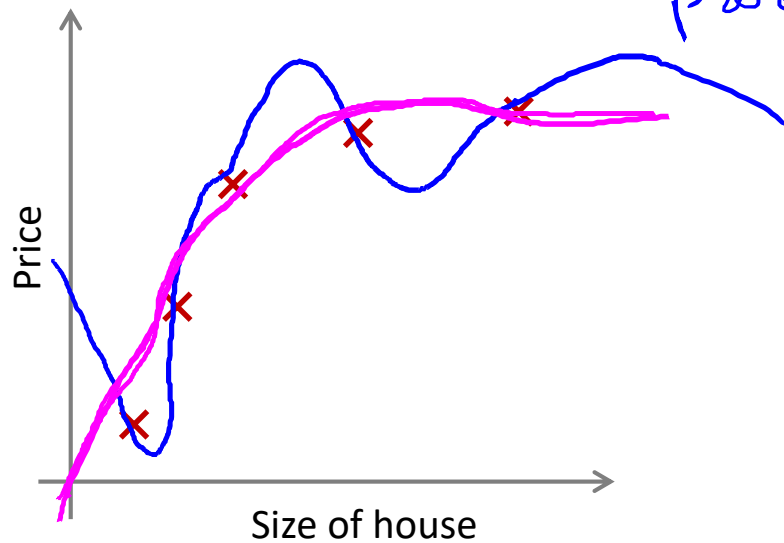
$$J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n \theta_j^2 \right]$$

~~θ_0~~ $\theta_1, \theta_2, \theta_3, \dots, \theta_{100}$ ~~θ_{100}~~

Regularization.

$$\rightarrow J(\theta) = \frac{1}{2m} \left[\underbrace{\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2}_{\text{blue arrow}} + \underbrace{\lambda \sum_{j=1}^n \theta_j^2}_{\text{pink arrow, regularization parameter}} \right]$$

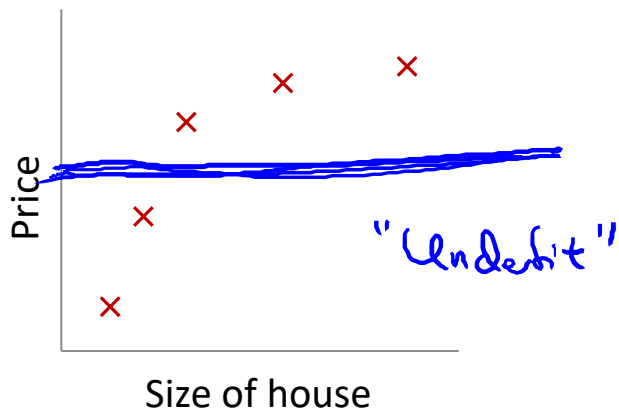
$\min_{\theta} J(\theta)$



In regularized linear regression, we choose θ to minimize

$$J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n \theta_j^2 \right]$$

What if λ is set to an extremely large value (perhaps far too large for our problem, say $\lambda = 10^{10}$)?



$h_{\theta}(x)$

$$\theta_0 + \cancel{\theta_1 x} + \cancel{\theta_2 x^2} + \cancel{\theta_3 x^3} + \cancel{\theta_4 x^4}$$

$\theta_1, \theta_2, \theta_3, \theta_4$

$\theta_1 \approx 0, \theta_2 \approx 0$

$\theta_3 \approx 0, \theta_4 \approx 0$

$$h_{\theta}(x) = \theta_0$$

Regularization

Regularized linear
regression

Regularized linear regression

$$J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \underbrace{\lambda \sum_{j=1}^n \theta_j^2} \right]$$

$$\min_{\theta} \underline{J(\theta)}$$

Gradient descent

$$\theta_0$$

↑

$$\theta_1, \theta_2, \dots, \theta_n$$

Repeat {

$$\frac{\partial}{\partial \theta_0} J(\theta)$$

$$\rightarrow \theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_0^{(i)}$$

$$\theta_j := \theta_j - \alpha$$

$$\frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

$$- \frac{\lambda}{m} \theta_j$$

$$(j = \cancel{0}, 1, 2, 3, \dots, n)$$

}

$$\rightarrow J(\theta)$$

$$\theta_j := \theta_j \left(1 - \alpha \frac{\lambda}{m}\right) - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

$$1 - \alpha \frac{\lambda}{m} < 1$$

$$0.99$$

$$\theta_j \times 0.99$$

$$\theta_j^2$$

Normal equation

$$\underline{X} = \begin{bmatrix} (x^{(1)})^T \\ \vdots \\ (x^{(m)})^T \end{bmatrix} \leftarrow$$

$m \times (n+1)$

$$\underset{\uparrow}{y} = \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(m)} \end{bmatrix} \quad \mathbb{R}^m$$

$$\rightarrow \min_{\theta} \underline{J(\theta)}$$

$$\Rightarrow \Theta = \left(X^T X + \lambda \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{(n+1) \times (n+1)} \right)^{-1} X^T y$$

$\frac{\partial}{\partial \theta_j} J(\theta) \stackrel{\text{set}}{=} 0$

$\in \mathbb{R}^n \quad n=2$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Non-invertibility (optional/advanced).

Suppose $m \leq n$, $\leftarrow$
(#examples) (#features)

$$\theta = (X^T X)^{-1} X^T y$$

$\underbrace{(X^T X)^{-1}}_{\text{non-invertible / singular}}$

pinv

inv
 $\nearrow$

If $\lambda > 0$,

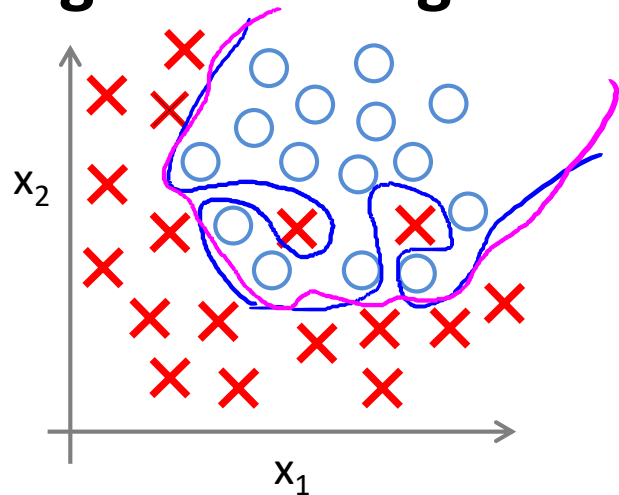
$$\theta = \left(X^T X + \lambda \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & \ddots \\ & & & & 1 \end{bmatrix} \right)^{-1} X^T y$$

$\underbrace{\hspace{10em}}_{\text{invertible}}$

Regularization

Regularized
logistic regression

Regularized logistic regression.



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^2 x_2 + \theta_4 x_1^2 x_2^2 + \theta_5 x_1^2 x_2^3 + \dots)$$

Cost function:

$$\rightarrow J(\theta) = - \left[\frac{1}{m} \sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right] + \frac{\lambda}{2m} \sum_{j=1}^n \theta_j^2$$

$\boxed{\theta_1, \theta_2, \dots, \theta_n}$

Gradient descent

Repeat {

$$\rightarrow \theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_0^{(i)}$$

$$\rightarrow \theta_j := \theta_j - \alpha \left[\underbrace{\frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}}_{\substack{(j = \cancel{0}, 1, 2, 3, \dots, n) \\ \theta_1, \dots, \theta_n}} - \frac{1}{n} \theta_j \right] \leftarrow$$

}

$$\frac{\partial}{\partial \theta_j} J(\theta)$$

$$\underline{h_{\theta}(x)} = \frac{1}{1 + e^{-\theta^T x}}$$